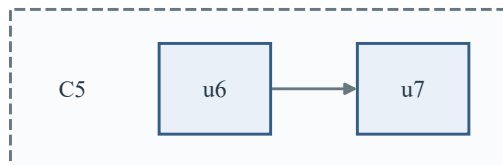
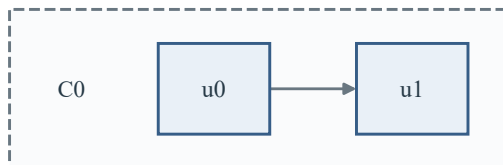
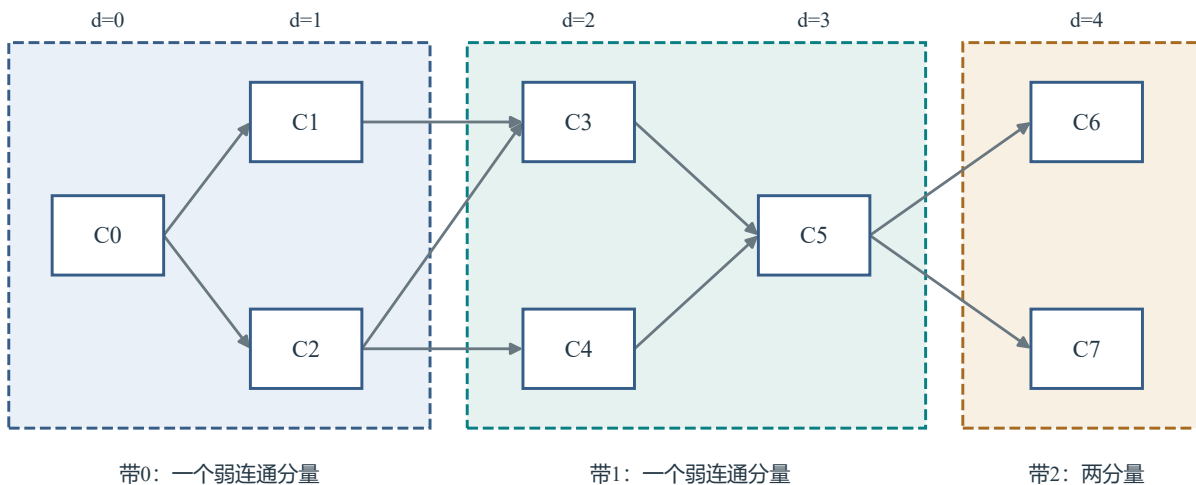


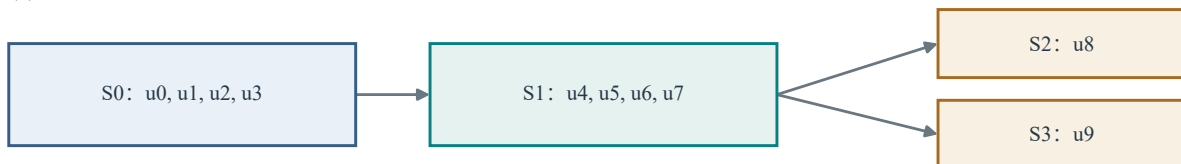
(a) 安全链：源出度=1，目标入度=1



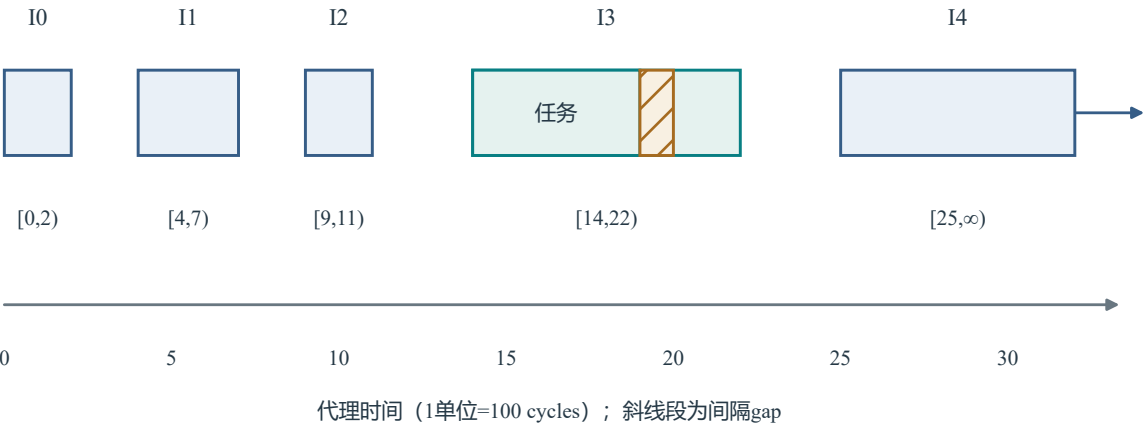
(b) 链块深度分带： $\text{band}(C) = \lfloor \text{depth}(C)/2 \rfloor$



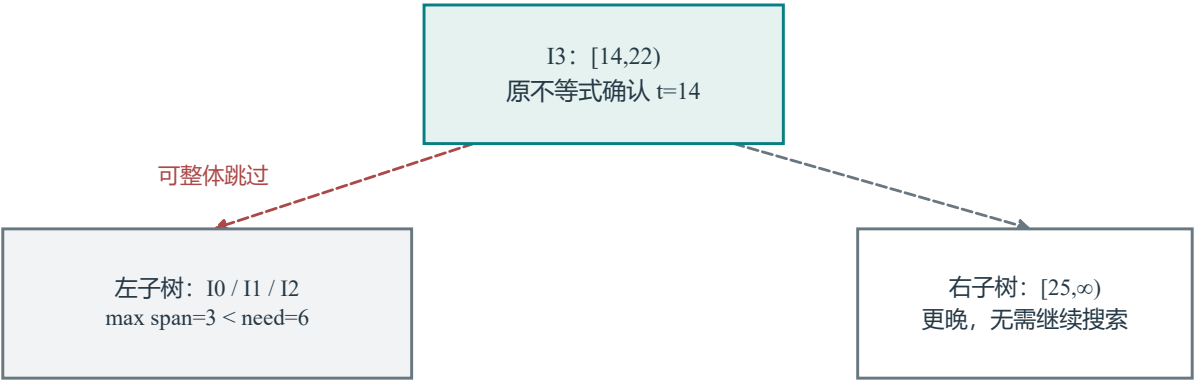
(c) 子图结果：带内按弱连通分量合并，不把整带强行并为一个Task



(a) 同一候选日历：ready=5，duration=5，gap=1



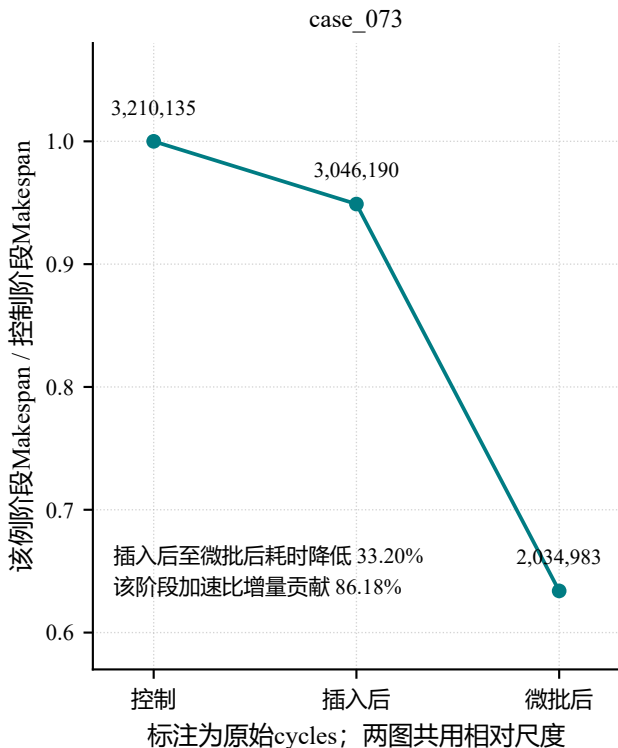
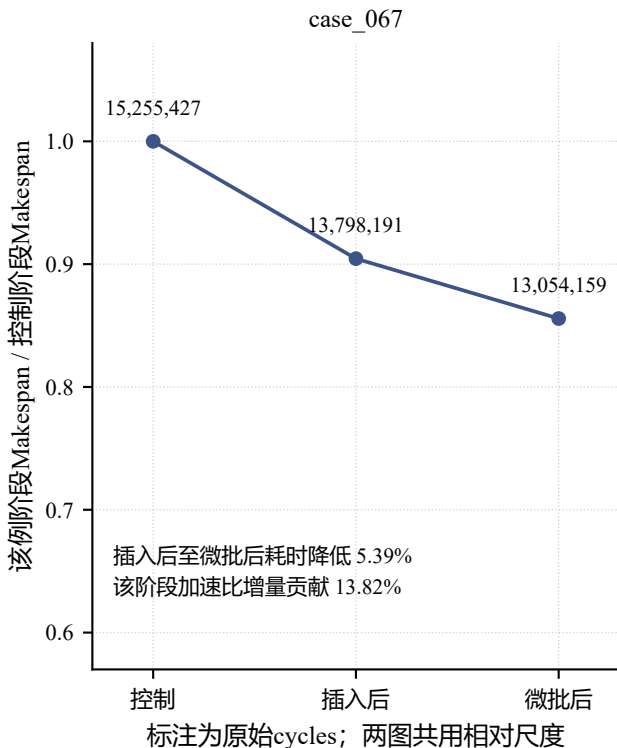
(b) 增强索引：用子树摘要排除不可能容纳任务的区间



$t = \max(\text{ready}, \text{low})$; $t + \text{duration} + \text{gap} \leq \text{high}$. t=14即1400代理cycles。

概念树形，非实际Treap快照；只索引候选日历，不改变官方时刻或评分。

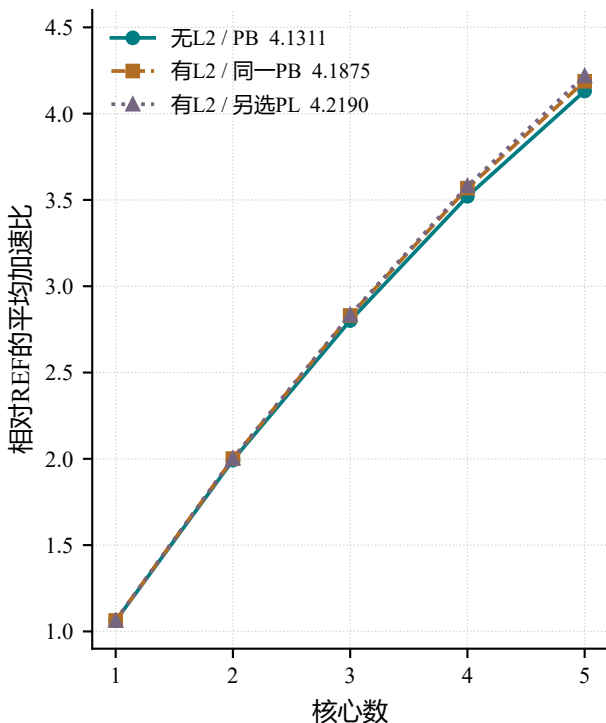
F21 微批阶段：两个实际改善案例的配对结果



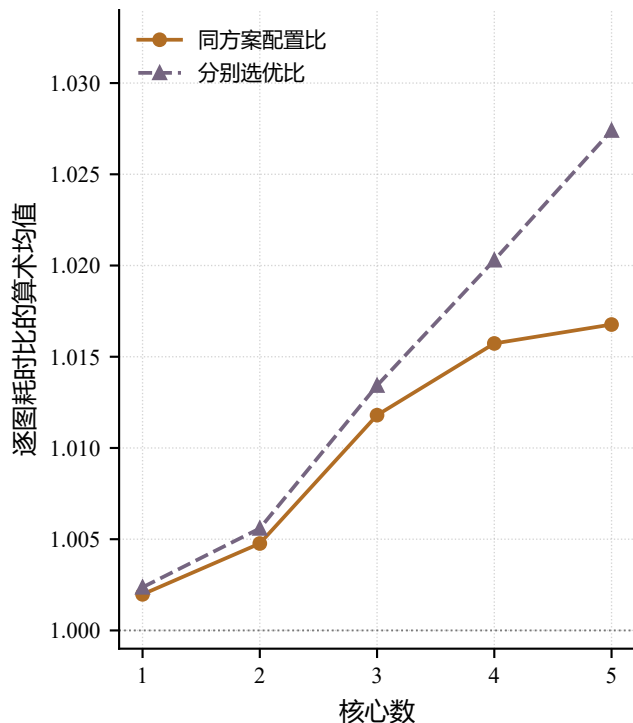
展示全部2个进一步改善例；其余98例未进一步改善，非等预算独立消融。

F24 L2: 三条方案曲线与两种配置比

(a) 三类真实方案；图例末列为五核值

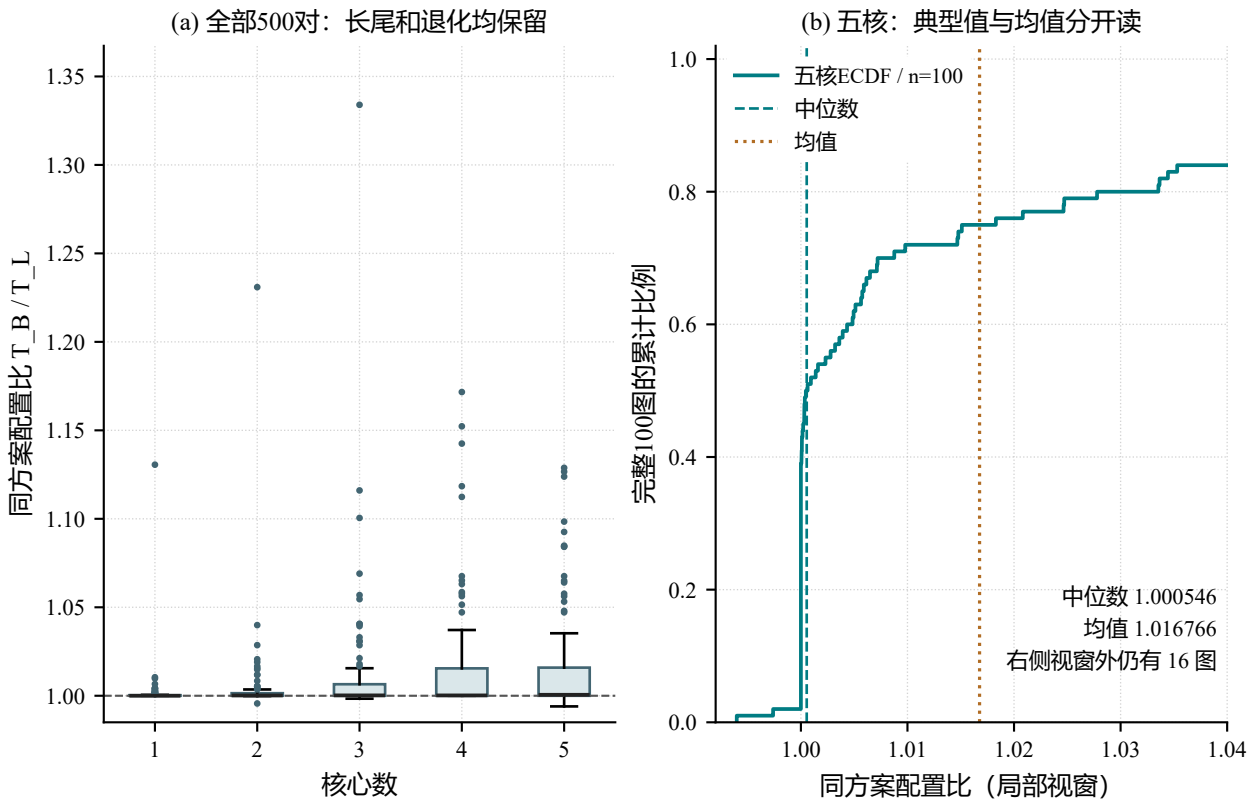


(b) 两种比值；纵轴局部放大



每点100图；右侧直接逐图求比再平均，不用左侧均值相除。

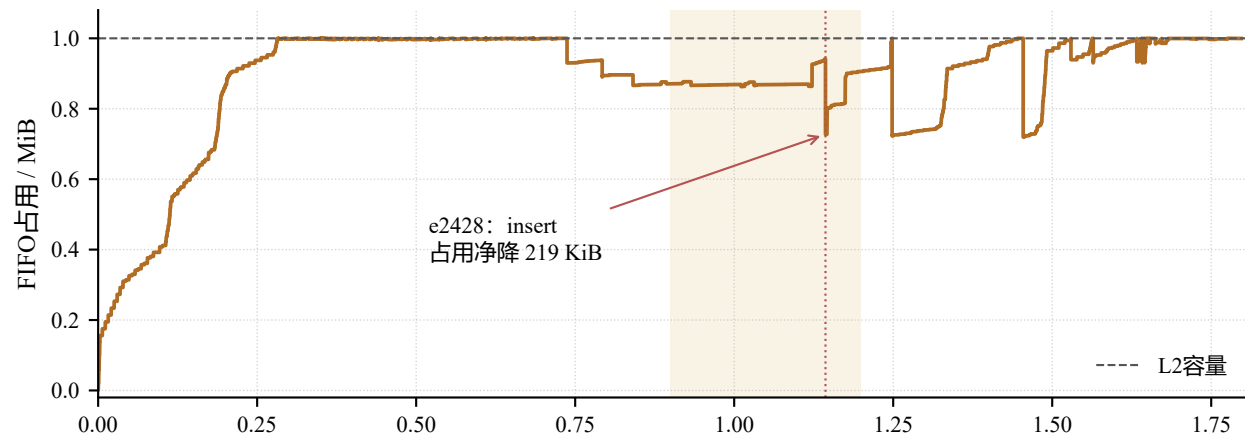
F25 L2收益：完整长尾与五核典型区间



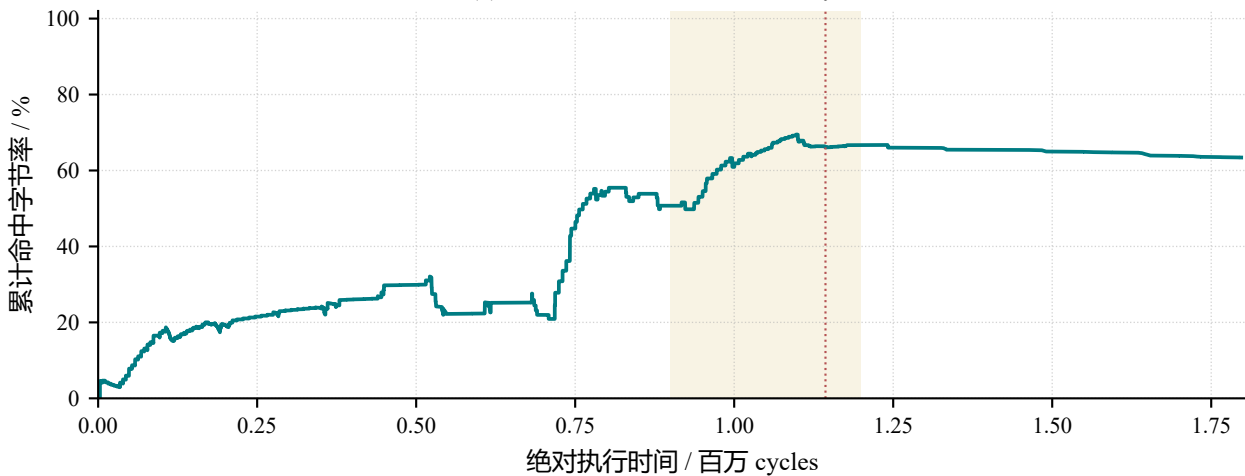
左图保留500对全范围；右图为五核100对ECDF的局部视窗，分母不变。

F27 case_073: 缓存事件与共同观察窗口

(a) 真实占用; 阴影与F28使用同一窗口

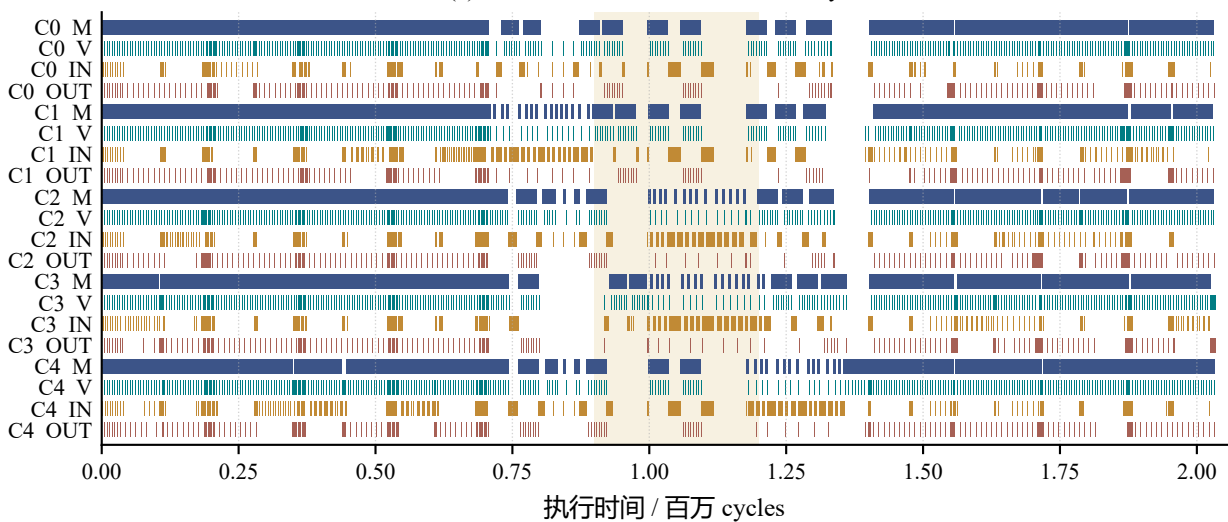


(b) 同一事件定位: $t=1,143,536$ cycles

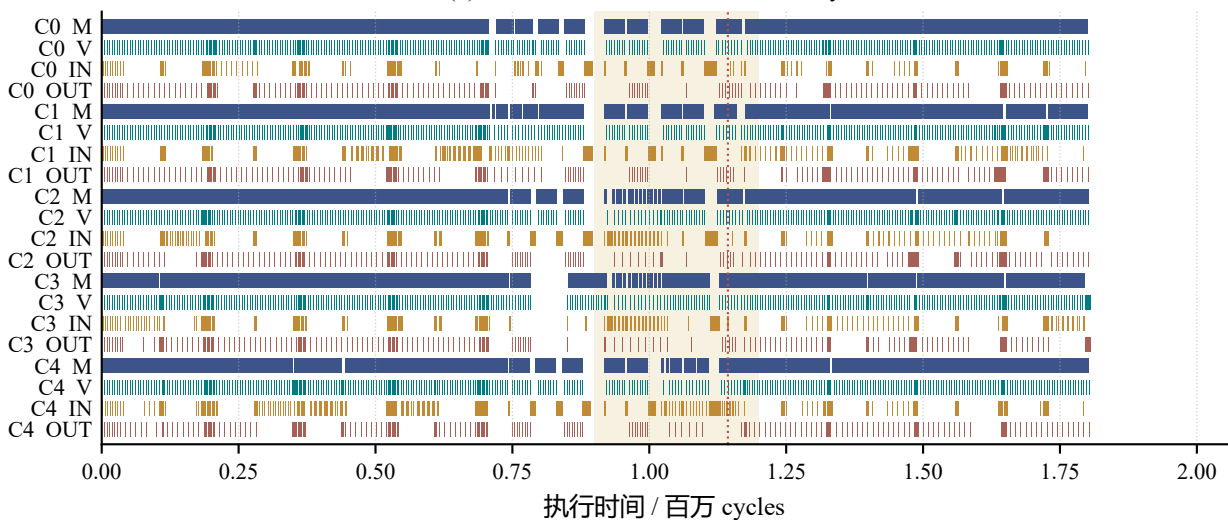


e为全局缓存日志序号; 净占用变化不等于驱逐字节, 更不是Pipe瓶颈归因。

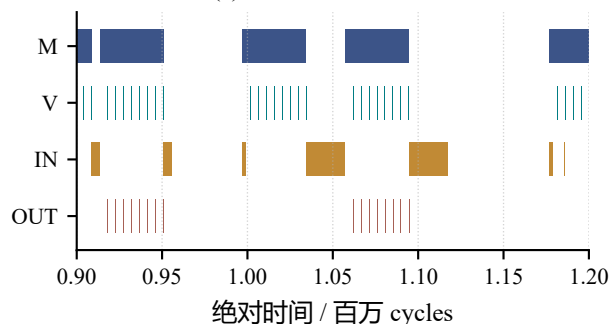
(a) 无L2: 固定PB; T = 2,034,983 cycles



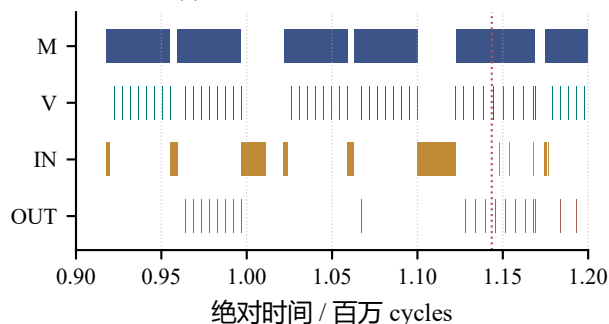
(b) 有L2: 同一PB; T = 1,806,253 cycles



(c) 无L2 / 核心0 / 局部



(d) L2 / 核心0; 红线e2428见F27



阴影为共同绝对时间窗; 红线是L2全局缓存事件, 不指定某Pipe的因果关系。